

Shesh Budhabhatti

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EDUCATION

NORTHEASTERN UNIVERSITY (Dean's List at Khoury College of Computer Sciences)	Boston, MA
Bachelor of Science in Cybersecurity and Business Administration (Finance)	April 2028
Coursework: Object-Oriented Design, Algorithms & Data Structures, CS Systems, Security Systems, Discrete Structures	

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL, Assembly
Systems & Tools: Git/GitHub, Linux/Unix, Docker, GDB, Valgrind, Azure DevSecOps
Security: Wireshark, Metasploit, Hashcat, threat modeling, penetration testing, cryptographic protocols
Spoken: English, Gujarati, Hindi, Spanish

WORK EXPERIENCE

MEDTRONIC	Boston, MA
Software Architecture Co-op	August 2026 – Present
- Building an automated pipeline that auto-generates UML diagrams and design docs from code updates for Medtronic's Hugo™ surgical robot platform - Unifying design docs, diagrams, and database records into one pipeline, migrating documentation to a core database with a cross-team API	
Product Security Intern	June 2026 – August 2026
- Built Python tooling to capture and visualize WiFi packet loss across 2.4/5 GHz bands for medical security research - Analyzed wireless attack surfaces and developed developer security training	
MICROSOFT	Boston, MA
AI Student Ambassador & WICxDevSecOps Program Participant	October 2025 - Feb 2026
- Selected for security training on Azure DevSecOps pipelines, threat modeling, and AI model security protocols - Demo Copilot/Omi AI to 100+ users, collecting feedback on quality, privacy risks, and edge cases for product teams	
MEDTRONIC	North Haven, CT
R&D Software Engineering Intern	Jun-Aug 2025 Jul-Aug 2024
- Reduced test fixture calibration time by 37% through Python automation scripts, saving 4+ hours weekly - Authored fixture manual for cross-functional teams across US/India (15+ engineers), reducing onboarding time - Implemented AES-256 and Blowfish encryption protocols for Signia™ Stapling System, ensuring HIPAA compliance and passing FDA cybersecurity audit requirements	
SECURE & ASSURED INTELLIGENT LEARNING LAB	New Haven, CT
Cybersecurity & Safety Researcher	Jun 2023 - Feb 2024
- Identified 8 security vulnerabilities in LLM systems through 50+ prompt injection attacks	
KPMG	Hartford, CT
Summer Intern, Audit/Tax/Advisory Accounting	July 2023 - August 2023
- Collaborated with 5-person teams across 3 companies, identifying 2 key control weaknesses in inventory tracking - Analyzed financial data across 3 client studies with Tableau, visualizing audit risk insight for senior stakeholders	
GURLS CHESS CLUB NON-PROFIT ORGANIZATION	USA
Founder & Executive Director & TEDx Speaker	December 2021 - Present
- Founded chess non-profit coaching 200+ students (70% female); delivered 70+ lessons, TEDx talk/USCF feature - Redesigned curriculum using retention data analysis, improving participant retention from 40% to 78%	

PROJECTS

Linux Security & Capture The Flag Challenges Penetration Testing, Linux	October 2025
- Achieved root access by exploiting SUID binaries and broken access controls across multi-level CTF challenges - Uncovered hidden files through HTML/JS inspection and steganography analysis, chaining system vulnerabilities to escalate privileges	
FUSE Filesystem Implementation C, Systems Programming	November 2025
- Built a custom filesystem supporting large files (>4KB) with inode management and mmap/munmap allocation - Debugged memory corruption using GDB and Valgrind, achieving 31/31 test cases.	
Memory Allocator & Data Structures C, Systems Programming	October 2025
- Implemented malloc/free using mmap/munmap with free list management and a first-fit allocation strategy - Optimized memory reuse through block coalescing, merging adjacent free blocks to minimize fragmentation	

AWARDS/PUBLICATIONS/CERTIFICATIONS

TEDx: "Empowering Equality, Inspiring Action" (2023) | SMARTA Podcast: "If You're Young, You Still Can Make a Difference" (2025) | Lean Six Sigma Yellow Belt (2024) | CT Women's Chess Champ. (2018) | TRAIN's Intro. to Artificial Intelligence Certified (2024)